



LAB 6

System Analysis and Design

OUR TEAM

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SAMPLING STRATEGY DESIGN

Target Populations:

- Regular users of the carbon footprint app (light, moderate, and heavy users)
- System-generated user activity logs

Selected Population: App users

Sampling Method: Stratified Random Sampling

JUSTIFICATION

Ensures balanced representation across user types
(light, moderate, heavy)

Reduces sampling bias

Makes findings more representative of the full user
base

Improves generalizability of results

Supports fairer, data-driven decisions

PRACTICAL STEPS

Segment users into 3 strata using activity logs.

Randomly select 5–6 users from each group.

Ensure demographic diversity within each stratum.

Contact selected users via email/in-app message.

Record sampling rationale for transparency and reproducibility.

INVESTIGATION OF EXISTING DATA

Potential Data Sources:

- User Action Logs (quantitative)
- App Store Reviews & Feedback (qualitative)
- Weekly Usage Reports (quantitative)

Selected Source: App Store Reviews & Feedback

Expected Insights:

- Common praise and issues (e.g., ease of use, speed)
- User expectations, confusion, and feature suggestions
- Input for feature prioritization and UX improvements

PLAYSCRIPT FOR OBSERVATION

Role: Giki Zero User

Scenario: Logging in to take carbon-reducing steps

Purpose: Identify info flow, UX gaps, and decision delays

Actor (User)	Script
Logs In	Enters email/password, checks carbon score
Explores Dashboard	Scans suggestions (e.g., energy, diet)
Selects an Action	Chooses "Reduce meat consumption," marks it complete
Reviews Impact	Sees updated footprint and Giki score
Shares Progress	Posts badge on social media for motivation

STROBE OBSERVATION PLAN

Context: A remote or hybrid professional using the app in a home office or co-working space

Insight: The setting supports sustainable habits and shows that users span beyond students including professionals and remote workers

Element	Observation/Insight
Location	Home or co-working desk; quiet, flexible setup
Desk Placement	Faces wall/window; promotes focus and openness
Stationery	Mostly digital tools; minimal paper use
Props	Reusable items (bottle, cup), plants - eco-conscious cues
External Sources	Blogs and forums alongside app
Lighting/Color	Warm LED lighting, natural tones - calm and efficient space
Clothing	Business casual - typical of remote work culture



SYNTHESIS WITH INTERACTIVE METHODS

#	Narrative	STROBE	Symbol	Interpretation
1	"I use the app during breaks."	Use in public/shared spaces	✓ Tick	Confirms casual, flexible usage across roles
2	"I get tips from blogs/socials too."	External content accessed	○ Circle	App use is supplemented—scope for content integration
3	"Hard to stay motivated."	No visible cues	👁️ Eye	Motivation issues need deeper methods (e.g., interviews)
4	"It fits my lifestyle."	Use aligns with context	■ Square	Environment shapes relevance perception

CONCLUSION

- Unobtrusive methods provide deep, authentic user insights
- They complement interactive tools for a user-centered design
- Findings support a more engaging, adaptive, and impact-driven platform



THANK YOU